CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry 41 (2013) 71–75

Contents lists available at SciVerse ScienceDirect

CALPHAD: Computer Coupling of Phase Diagrams and Thermochemistry

journal homepage: www.elsevier.com/locate/calphad

Obituary

Thermodynamic assessment of the Sc–Sn system

ARTICLE INFO

Keywords:
Sc–Sn system
Phase diagram
Thermodynamic assessment
CALPHAD method

ABSTRACT

The thermodynamic optimization of the Sc–Sn binary system was carried out with the help of CALculation of PHAse Diagram (CALPHAD) method. Sc_5Sn_3, Sc_6Sn_5, $Sc_{11}Sn_{10}$ and $ScSn_2$ have been treated as stoichiometric compounds while a solution model has been used for the description of the liquid, HCP_A3 (α-Sc) and BCC_A2 (β-Sc) phases. The calculations based on the thermodynamic modeling are in good agreement with the phase diagram data and experimental thermodynamic values.

1. Introduction

The alloy systems or intermetallic compounds composed of rare earth and Sn have attracted much attention for the theoretical study of the solid state and for many industrial purposes.

In this work, the binary Sc–Sn system was optimized using the CALPHAD technique [1]. The thermodynamic parameters involved in the models are optimized from the experimental thermodynamic and phase diagram data.

2. Review of experimental data

The phase equilibria of the Sc–Sn binary system were firstly measured by Palenzona and Manfrinetti [2] using thermal analysis technique in the range 30–100 at%. Four intermetallic compounds were reported: Sc_5Sn_3, Sc_6Sn_5, ScSn and $ScSn_2$. For the Sc_5Sn_3 compound a congruent melting was determined at 2073 K, while the Sc_6Sn_5, ScSn and $ScSn_2$ compounds, a peritectic formation was reported, respectively, at 1728, 1218 and 1168 K.

Later, by X-ray diffraction and microscopy technique, Podarevskaya et al. [3] reinvestigated the phase diagram in the range 0–60 at%. Three intermetallic compounds, Sc_5Sn_3, Sc_5Sn_4 and $Sc_{11}Sn_{10}$ were identified. For the Sc_5Sn_3 compound a congruent melting was described at 2073 K, while the Sc_5Sn_4 and $Sc_{11}Sn_{10}$ compounds, a peritectic formation were reported, respectively, at 1643 and 1238 K.

A polymorphic transition α-Sc_5Sn_3 ↔ β-Sc_5Sn_3 was determined at 1988 K for Sc_5Sn_3.

Recently, the evaluated Sc–Sn phase diagram by Okamoto [4] (Fig. 1) is based on the work of Palenzona and Manfrinetti [2] and Podarevskaya et al. [3] according to this assessment. For the Sc_5Sn_3 compound a congruent melting was estimated at 2073 K, Sc_6Sn_5, $Sc_{11}Sn_{10}$ and $ScSn_2$ compounds, a peritectic formation were reported respectively at 1643, 1238 and 1148 K. A polymorphic transition α-Sc_5Sn_3 ↔ β-Sc_5Sn_3 was determined at 1988 K for Sc_5Sn_3. The crystal structures of various phases are reported in Table 1.

Podarevskaya [5] determined the enthalpies of formation for three intermetallic compounds Sc_5Sn_3, Sc_6Sn_5 and ScSn. Lebedev et al. [6] determined the enthalpies of formation of the $ScSn_2$. Meschel and Kleppa [7] measured the standard enthalpy of formation of Sc_5Sn_3 using the dynamic differential calorimetry method.

3. Thermodynamic models

3.1. Pure elements

The Gibbs energy of the pure element i (i=Sc, Sn) in the phase φ (φ=Liquid, HCP_A3, BCT_A5 and BCC_A2), referred to the enthalpy of its stable state at 298.15 K, is described as a function of temperature by

$$G_i^{\varphi}(T) = {}^0G_i^{\varphi} - H_i^{SER}(298.15\ \mathrm{K}) = a + bT + cTlnT + dT^2 + eT^3 + fT^7 + gT^{-1} + hT^{-9} \quad (1)$$

where H_i^{SER}(298.15 K) is the molar enthalpy of the element i at 298.15 K in its standard element reference (SER) state, HCP_A3 for Sc and BCT_A5 for Sn.

In this article, the Gibbs energy functions are taken from the SGTE compilation of Dinsdale [8].

3.2. Solution phases

The solution phases [(α-Sc), (β-Sc) and liquid] were modeled as substitutional solutions according to the polynomial Redlich–Kister model [9]. The solid BCT_A5 (β-Sn) and Diamond_A4 (α-Sn) with negligible solubilities are considered as pure elements. The Gibbs energy of one mol of formula unit of phase φ is expressed as the sum of the reference part ${}^{ref}G^{\varphi}$, the ideal part ${}^{id}G^{\varphi}$, and the

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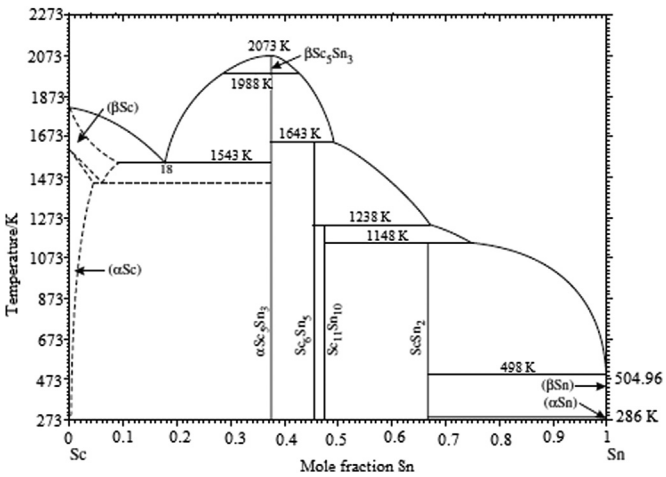

Fig. 1. The phase diagram of the Sc–Sn system reviewed by Okamoto [4].

Table 1
Symbols and crystal structures of the stable solid phases in the Sc–Sn system.

Diagram symbol	Composition Pearson at% Sn	symbol	Symbol used in Thermo-Calc data file	Prototype	Reference
β-Sc	0	cI2	BCC_A2	W	[3]
α-Sc	0	hP2	HCP_A3	Mg	[3]
β-Sc_5Sn_3	37.5	–	Sc_5Sn_3_ht	–	[3]
α-Sc_5Sn_3	37.5	hP16	Sc_5Sn_3_Bt	Mn_5Si_3	[2,3]
Sc_6Sn_5	45.5	oI44	Sc_6Sn_5	Ti_6Ge_5	[2]
$Sc_{11}Sn_{10}$	47.6	–	$Sc_{11}Sn_{10}$	–	[3]
$ScSn_2$	66.7	t124	$ScSn_2$	–	[2]
β-Sn	100	tI4	BCT_A5	β-Sn	[2]
α-Sn	100	cF8	Diamond_A4	C	[2]

excess part ${}^{\mathrm{exc}}G^{\varphi}$:

$$G_m^{\varphi} = {}^{\mathrm{ref}}G^{\varphi} + {}^{\mathrm{id}}G^{\varphi} + {}^{\mathrm{exc}}G^{\varphi} \tag{2}$$

As used in the Thermo-Calc software [10]:

$${}^{\mathrm{ref}}G^{\varphi}(T) = ({}^{0}G_{\mathrm{Sc}}^{\varphi}(T) - H_{\mathrm{Sc}}^{\mathrm{SER}}(298.15\ \mathrm{K}))x_{\mathrm{Sc}} + ({}^{0}G_{\mathrm{Sc}}^{\varphi}(T) - H_{\mathrm{Sn}}^{\mathrm{SER}}(298.15\ \mathrm{K}))x_{\mathrm{Sn}} \tag{3}$$

$${}^{\mathrm{id}}G^{\varphi} = RT(x_{\mathrm{Sc}}\ln x_{\mathrm{Sc}} + x_{\mathrm{Sn}}\ln x_{\mathrm{Sn}}) \tag{4}$$

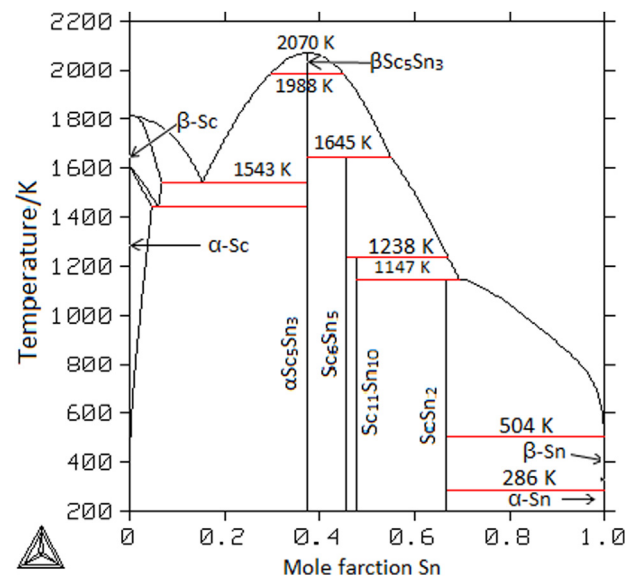

Fig. 2. Calculated Sc–Sn phase diagram.

Table 2
The optimized thermodynamic parameters of the Sc–Sn system.

phase	Thermodynamic models	Parameters in SI
Liquid	$(Sc,Sn)_1$	${}^{0}L^{\mathrm{liq}} = -144{,}436 - 9.409T$ ${}^{1}L^{\mathrm{liq}} = -24{,}148 + 6.112T$ ${}^{2}L^{\mathrm{liq}} = 178 + 30.207T$
BCC_A2	$(Sc,Sn)_1(Va)_3$	${}^{0}L^{\mathrm{BCC_A2}} = -131{,}962 - 0.206T$
HCP_A3	$(Sc,Sn)_1(Va)_{0.5}$	${}^{0}L^{\mathrm{HCP_A3}} = -127{,}264 + 0.002T$
BCT_A5	(Sn)	No excess term
DIAMOND_A4	(Sn)	No excess term
$ScSn_2$	$(Sc)_{0.333}$: $(Sn)_{0.667}$	$G_{\mathrm{Sc:Sn}}^{\mathrm{ScSn_2}} - 0.333\,{}^{298}H_{\mathrm{Sc}}^{\mathrm{HCP_A3}} - 0.667\,{}^{298}H_{\mathrm{Sn}}^{\mathrm{BCT_A5}}$ $= 0.333G_{\mathrm{Sc}}^{\mathrm{HCP_A3}} + 0.667G_{\mathrm{Sn}}^{\mathrm{BCT_A5}} - 44{,}922 + 2.106T$
$Sc_{11}Sn_{10}$	$(Sc)_{0.523}$: $(Sn)_{0.477}$	$G_{\mathrm{Sc:Sn}}^{\mathrm{Sc_{11}Sn_{10}}} - 0.523\,{}^{298}H_{\mathrm{Sc}}^{\mathrm{HCP_A3}} - 0.477\,{}^{298}H_{\mathrm{Sn}}^{\mathrm{BCT_A5}}$ $= 0.523G_{\mathrm{Sc}}^{\mathrm{HCP_A3}} + 0.477G_{\mathrm{Sn}}^{\mathrm{BCT_A5}} - 52{,}720 - 2.261T$
Sc_6Sn_5	$(Sc)_{0.545}$: $(Sn)_{0.455}$	$G_{\mathrm{Sc:Sn}}^{\mathrm{Sc_6Sn_5}} - 0.545\,{}^{298}H_{\mathrm{Sc}}^{\mathrm{HCP_A3}} - 0.455\,{}^{298}H_{\mathrm{Sn}}^{\mathrm{BCT_A5}}$ $= 0.545G_{\mathrm{Sc}}^{\mathrm{HCP_A3}} + 0.455G_{\mathrm{Sn}}^{\mathrm{BCT_A5}} - 53{,}661 - 2.568T$
α-Sc_5Sn_3	$(Sc)_{0.625}$: $(Sn)_{0.375}$	$G_{\mathrm{Sc:Sn}}^{\alpha\mathrm{Sc_5Sn_3}} - 0.625\,{}^{298}H_{\mathrm{Sc}}^{\mathrm{HCP_A3}} - 0.375\,{}^{298}H_{\mathrm{Sn}}^{\mathrm{BCT_A5}}$ $= 0.625G_{\mathrm{Sc}}^{\mathrm{HCP_A3}} + 0.375G_{\mathrm{Sn}}^{\mathrm{BCT_A5}} - 56{,}020 - 1.777T$
β-Sc_5Sn_3	$(Sc)_{0.625}$: $(Sn)_{0.375}$	$G_{\mathrm{Sc:Sn}}^{\beta\mathrm{Sc_5Sn_3}} - 0.625\,{}^{298}H_{\mathrm{Sc}}^{\mathrm{HCP_A3}} - 0.375\,{}^{298}H_{\mathrm{Sn}}^{\mathrm{BCT_A5}}$ $= 0.625G_{\mathrm{Sc}}^{\mathrm{HCP_A3}} + 0.375G_{\mathrm{Sn}}^{\mathrm{BCT_A5}} - 55{,}997 - 1.789T$

where R is the gas constant, T the temperature, in Kelvin, x_{Sc} and x_{Sn} are the fraction of elements Sc and Sn, respectively.

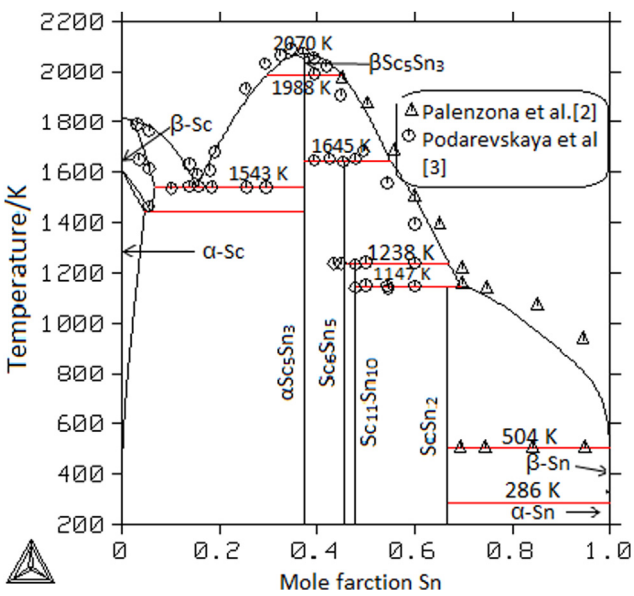

Fig. 3. Comparison of the Sc–Sn calculated phase diagram with the experimental data.

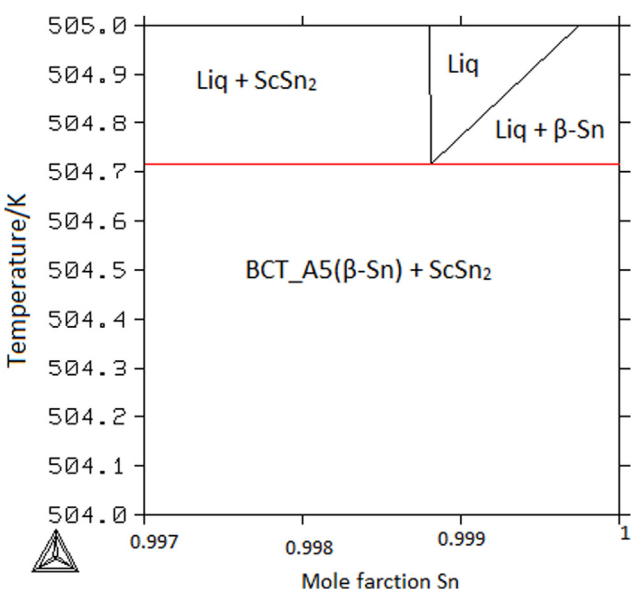

Fig. 4. Enlarged part on the Sn-rich Sc–Sn eutectic region.

The excess terms of all the solution phases were modeled by the Redlich–Kister [9] formula:

$$^{exc}G_m^{\varphi}(T) = x_{Sc}x_{Sn}[^0L_{Sc,Sn}^{\varphi}(T) + {}^1L_{Sc,Sn}^{\varphi}(T)(x_{Sc}-x_{Sn}) + {}^2L_{Sc,Sn}^{\varphi}(T)(x_{Sc}-x_{Sn})^2 + \cdots] \quad (5)$$

$$^iL_{Sc,Sn}^{\varphi}(T) = a_i + b_iT \quad (6)$$

where $^iL_{Sc,Sn}^{\varphi}(T)$ is the interaction parameter between the elements Sc and Sn, which is evaluated in the presented work, a_i and b_i are the coefficients to be optimized.

3.3. Stoichiometric compounds

The Gibbs energy of the stoichiometric compounds ApBq is expressed as follows:

$$^0G_{A_pB_q} = \frac{p}{p+q}\,{}^0G_A + \frac{q}{p+q}\,{}^0G_B + a + bT \quad (7)$$

where ^0G_A and ^0G_B are the Gibbs energy of the pure elements Sc and Sn, respectively, a and b are parameters to be determined.

4. Results and discussions

The thermodynamic parameters for different phases were evaluated with the PARROT [11] module in Thermo-Calc software [10], which works by minimizing the square sum of the differences between measured and calculated values. The liquid was treated as a substitutional solution. For all the intermetallic compounds (Sc_5Sn_3, Sc_6Sn_5, $Sc_{11}Sn_{10}$ and $ScSn_2$) were taken into account and treated as stoichiometric compounds. The (α-Sc_5Sn_3 ↔ β-Sc_5Sn_3) polymorphic transformation was considered. Most of the experimental data mentioned in Section 2 were employed in the optimization. The thermodynamic parameters of Eq. (7) were adjusted according to the enthalpies of formation measured by [5–7]. The compositions of the phases and the temperatures involved in the invariant reactions mainly based on the phase diagram of the Sc–Sn system evaluated by Okamoto [4] were used and all the thermodynamic parameters were optimized and listed in Table 2.

The calculated phase diagram is shown in Fig. 2. It is compared with the numerous experimental data in Fig. 3. A good agreement is noted.

The Sn-rich region of the Sc–Sn phase diagram is plotted in Fig. 4. The temperature of this Liq ↔ Sn + $ScSn_2$ eutectic reaction is calculated at 504 K.

All the invariant equilibria in the Sc–Sn system are listed in Table 3. The calculated phase diagram is in excellent agreement with the most of experimental result reported by Palenzona and Manfrinetti [2] and Podarevskaya et al. [3]. But the compositions of Sn liquid at the peritectic temperatures of 1643 K have a relative larger discrepancy between calculated value and experimental data of Podarevskaya et al. [3].

Table 3
Invariant reactions in the Sc–Sn system.

Reaction	Palenzona and Manfrinetti [2]		Podarevskaya et al. [3]		This work	
	T (K)	x_{liq} (at% Sn)	T (K)	x_{liq} (at% Sn)	T (K)	x_{liq} (at% Sn)
Liq ↔ Sn + $ScSn_2$	498	0.99	–	–	504.7 ± 3	0.99 ± 0.01
Liq + $ScSn_2$ ↔ $Sc_{11}Sn_{10}$	1163	0.73	1148 ± 5	–	1147 ± 3	0.69 ± 0.01
Liq + $Sc_{11}Sn_{10}$ ↔ Sc_6Sn_5	1218	0.67	1238 ± 5	–	1238 ± 2	0.66 ± 0.01
Liq + Sc_6Sn_5 ↔ α-Sc_5Sn_3	1728	0.53	1643 ± 10	0.49	1645 ± 3	0.54 ± 0.01
α-Sc_5Sn_3 ↔ β-Sc_5Sn_3	–	–	1988 ± 10	–	1988 ± 2	–
Liq ↔ β-Sc_5Sn_3	2073	–	2073	–	2070 ± 5	–
Liq ↔ Sc + α-Sc_5Sn_3	–	–	1543 ± 5	0.18	1543 ± 2	0.15 ± 0.01

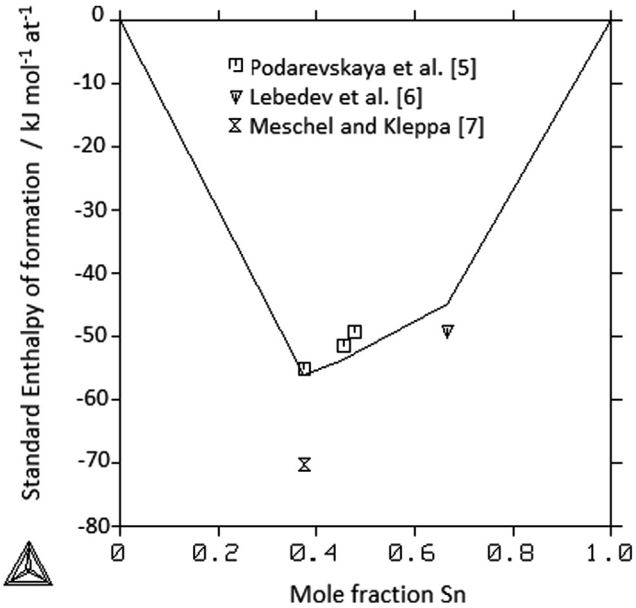

Fig. 5. Calculated and measured enthalpies of formation of the intermetallic compounds.

Table 4
Calculated and measured enthalpies of formation of the intermetallic compounds.

	Enthalpy of formation/kJ mol^{-1} at^{-1}			
	This work	Podarevskaya [5]	Lebedev et al. [6]	Meschel and Kleppa [7]
$ScSn_2$	−44.922		−48.9 ± 5.7	
$Sc_{11}Sn_{10}$	−52.720	−49.3 ± 1.5		
Sc_6Sn_5	−53.661	−51.5 ± 1.4		
α-Sc_5Sn_3	−56.020	−55.2 ± 1.2		−70.2 ± 1.8
β-Sc_5Sn_3	−55.997			

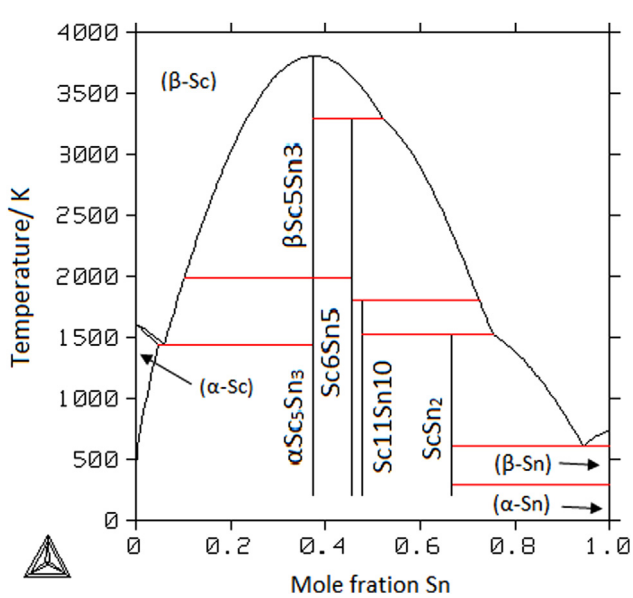

Fig.6. Calculated Sc–Sn phase diagram when the liquid phase is suspended.

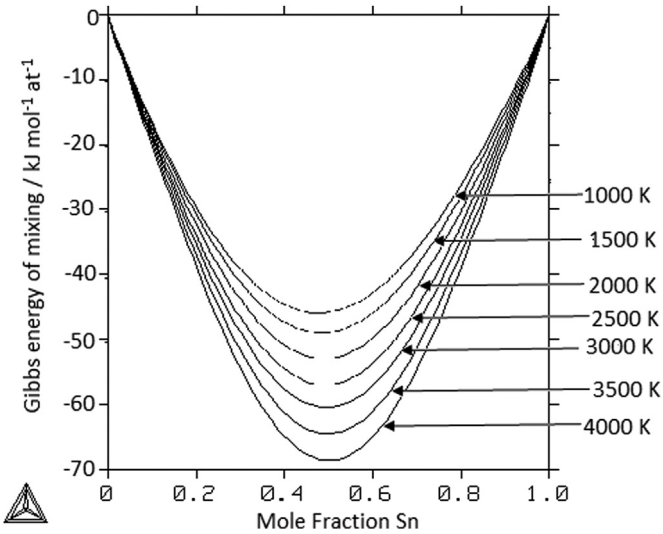

Fig. 7. Gibbs energy of mixing of the liquid phase at different temperatures. The reference states were Sc and Sn liquids.

The assessed enthalpies of formation of the intermetallic compounds compared with experimental measurements are plotted in Fig. 5 and Table 4. The calculated enthalpies agree well with the experimental data [5,7].

As mentioned by [12], in order to check that the optimized thermodynamic parameters of the intermetallic compounds are satisfactory, we verified that when the liquid phase is suspended during the calculation of the Sc–Sn phase diagram, the stoichiometric phases disappear at high temperatures.

A terminal solid solutions and a two-phase domain existing between them are found to be stable (Fig. 6).

Finally, we verified that no miscibility gap was calculated with our thermodynamic optimized parameters as commended in [13]. Fig. 7 shows the evolutions of Gibbs energy for the liquid phase as a function of temperature (T), when T is increasing up to 4000 K, the Gibbs energy for the liquid phase decreases. No anomaly is detected on all the curves.

5. Conclusions

On the basis of the critical review of the experimental information on phase diagram and thermodynamic properties in the published literature, thermodynamic assessment of the Sc–Sn binary system has been performed using the CALPHAD method. The calculated phase equilibria agree with the literature data. However, this description should be viewed as a preliminary one. More experimental work on this system may be necessary to improve the description.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2013.01.006.

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Received 16 July 2012

Available online 22 February 2013

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